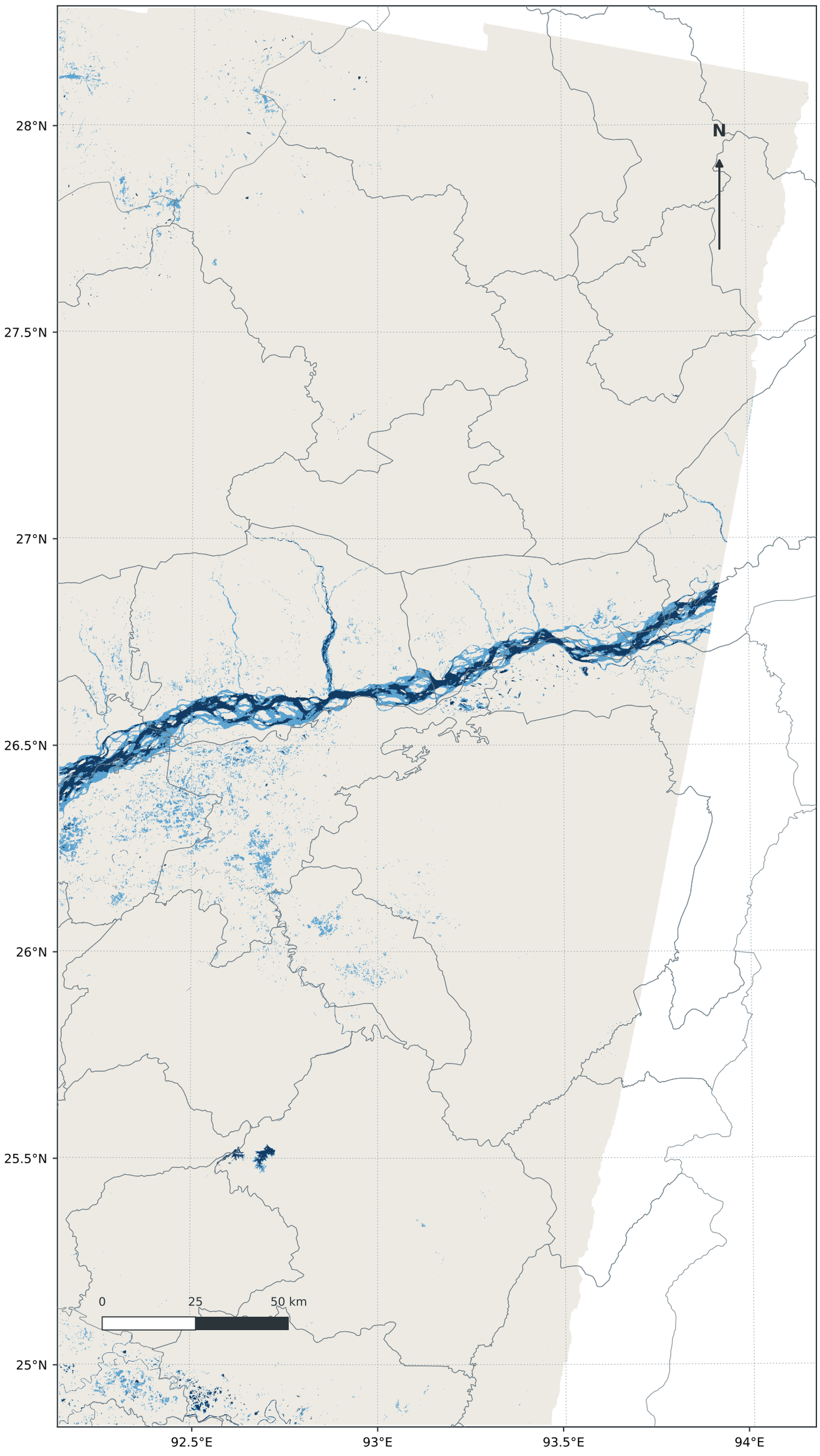
Brahmaputra valley flood extent and flooded cropland

Sentinel-1 SAR, 12 August 2016 · monsoon season, Kharif 2016 · central Brahmaputra valley, Assam and adjoining states

Flood extent, 12 August 2016



LEGEND

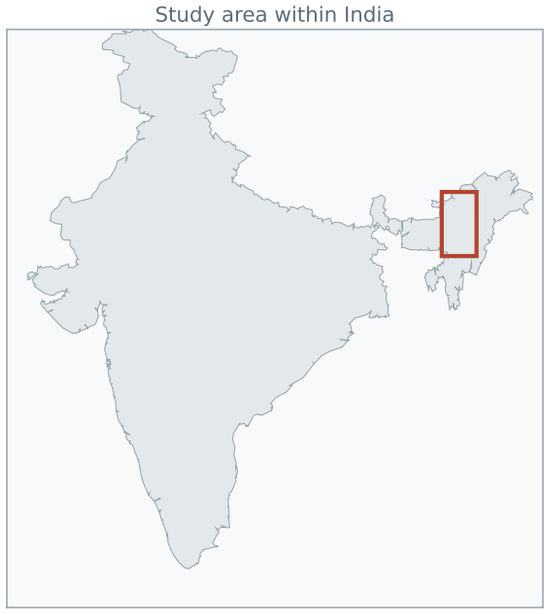
- Permanent water — *JRC occurrence ≥ 50%*
- Flood water — *dark on 12 Aug, not permanent*
- Land, imaged
- No data — *outside swath, or edge buffer*

AREA BY CLASS

class	hectares	operating-point range
Flood water	**119,779**	119,779 to 170,625
Flooded cropland	**46,943**	46,943 to 87,127
Permanent water	58,389	58,389 to 60,076
Land, imaged	6,034,074	6,034,074 to 6,086,607
No data	1,484,816	—

Range spans two defensible thresholds: −18.75 dB, maximum agreement with hand labels, and −17.75 dB, area matching them before clean-up, about a tenth low after it. Not a confidence interval.

Study area within India



PROJECTION EPSG:32646 (UTM zone 46N), 20 m pixel

SENSOR Sentinel-1A IW GRD, RTC gamma0, VH polarisation

ACQUISITION 12 August 2016, 23:46 UTC, relative orbit 77, descending

METHOD Lee 5×5 speckle filter in linear power, single-date threshold, 640 m swath-edge buffer to no data, slope > 8° reclassified as land, minimum mapping unit 0.4 ha

PERMANENT WATER JRC Global Surface Water occurrence ≥ 50% (Pekel et al. 2016)

CROPLAND ESA WorldCover 2021 class 40. Four to five years after the event; no open 10 m cropland map exists for 2016

BOUNDARIES geoBoundaries ADM2 India, current vintage, CC BY 4.0. Assam districts have been created since 2016

VALIDATION Sen1Floods11 hand labels, 65 chips, IoU 0.519 (95% CI [0.395, 0.609]). The refinement steps above are unvalidated: no labelled data covers them

PRODUCED 13 October 2026 · github.com/sruthi-swathandran/geospatial-portfolio

Contains modified Copernicus Sentinel data 2016, processed by Microsoft Planetary Computer. Land cover © ESA WorldCover 2021. Surface water: JRC/Google. Boundaries: geoBoundaries.